

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2009

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

- 4.** Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2009.

1. Which one of the following components of blood is involved in clotting after an injury?
 - A. Plasma
 - B. White blood cells
 - C. Red blood cells
 - D. Platelets
2. Which one of the following shows the CORRECT order of blood flow in the heart?
 - A. Vena cava -> pulmonary artery -> pulmonary vein -> aorta
 - B. Pulmonary artery -> vena cava -> pulmonary vein -> aorta
 - C. Pulmonary vein -> aorta -> pulmonary artery -> vena cava
 - D. Aorta -> pulmonary vein -> vena cava -> pulmonary artery
3. Which of the following pairs consists only of social effects of drug abuse?
 - A. Rape and truancy
 - B. Lack of concentration and addiction
 - C. Accidents and withdrawal
 - D. Impaired judgment and loss of consciousness
4. Which one of the following statements is TRUE about the incubation stage of HIV infection?
 - A. If HIV test is carried out the results is negative
 - B. This is the first stage of infection
 - C. There are no visible signs
 - D. The body cannot fight diseases
5. Which one of the following is NOT a CORRECT statement about foetus
 - A. Receives oxygen through placenta
 - B. The blood of the mother and that of the foetus mix
 - C. Receives food substances from the mother's blood
 - D. Waste products are removed through the placenta
6. Which one of the following pairs of infections consists only of those that are sexually transmitted?
 - A. Chancroid and HIV
 - B. Gonorrhea and bilharzia
 - C. Syphilis and typhoid

- D. Gonorrhea and tuberculosis
7. A ruler placed slanting and partially immersed in water appears bent because light,
- A. From that part of the ruler under water is reflected
 - B. From that part of the ruler under water is refracted
 - C. From that part of the ruler under water is scattered
 - D. Travels in the straight line in the same medium
8. The diagram below represents a set up that can be used to demonstrate a certain aspect of heat
- The aspect demonstrated is that
- A. Solids melt when heated
 - B. Metals expand when heated
 - C. Solids conduct heat
 - D. Metals bend when heated
9. Which one of the following is storage pest?
- A. Weaver birds
 - B. White ant
 - C. Cutworm
 - D. Aphid
10. Which one of the following stores food in the same part of the plants as an Irish potato?
- A. Carrot
 - B. Sweet potato
 - C. sugarcane
 - D. cassava
11. The embryo of seed is made up of
- A. plumule and testa
 - B. cotyledon and plumule
 - C. radicle and plumule
 - D. cotyledon and radicle
12. The following are some characteristics of certain animals;
- i) Body covered with scale
 - ii) Lay eggs
 - iii) Body temperature varies with the surroundings
 - iv) Breathe through lungs

Which one of the following pairs of animals has ALL the characteristics listed?

- A. Turtle and shark
- B. Lizard and tortoise
- C. Tilapia and frog
- D. Crocodile and platypus

13. Which one of the following is the main constituent of a balanced diet obtained by animals when they feed on lucern?
- A. Carbohydrates
 - B. Protein
 - C. Vitamins
 - D. Minerals
14. Which one of the following is NOT an example of rotational grazing?
- A. Strip
 - B. Paddockding
 - C. Tethering
 - D. Herding
15. The following are activities carried out when modeling the solar system but not in their correct order
- i) Draw circles on manila paper to show orbit
 - ii) Model the planets and the sun using clay
 - iii) Put name tags on models of planets and the sun
 - iv) Fix models of planets to orbit and the model of the sun at the centre
 - v) Paste manila paper on the soft board
- Which one of the following is the CORRECT order in which the activities are carried out?
- A. (i), (ii), (iv) (v)
 - B. (i) (iii) (ii) (v) (iv)
 - C. (ii) (iv) (iii) (i) (v)
 - D. (ii) (v) (i) (iv) (iii)
16. The following are some uses of components of air
- i) Manufacture of electric bulbs
 - ii) Supporting burning
 - iii) Manufacture of soft drinks
 - iv) Germination of seed
 - v) Putting out fires
- Which of the uses are for carbon dioxide?
- A. (i) and (ii)
 - B. (ii) and (iv)
 - C. (iii) and (v)
 - D. (iii) and (iv)
17. A certain machine when in use can work with load effort and fulcrum at different position as shown in the diagram I and II
- The machine is a
- A. Wheelbarrow
 - B. Crowbar

- C. Spade
- D. Claw hammer

18. The following are methods of separating mixture;

- i) Picking
- ii) Filtering
- iii) Decanting
- iv) Sieving

Which two methods are suitable for separating a mixture of water and sand?

- A. (I) and (iii)
- B. (i) and (iv)
- C. (ii) and (iii)
- D. ?(ii) and (iv)

19. Which one of the following is a CORRECT food chain?

- A. Grass-rabbit-eagle-leopard
- B. Maize-grasshopper-lizard-snake
- C. Grass-grasshopper-eagle-rabbit
- D. Kales-aphids-weaverbirds-rabbit

20. When pieces of paper are dropped above a fire they rise. This is because

- A. Heat makes the paper lighter than air
- B. Paper expand when heated
- C. Hot air carries the paper upwards
- D. Heat from the fire carries the paper upwards

21. The diagram below represents a set up that was used to demonstrate a certain property of matter

The property demonstrated was

- A. Air occupies space
- B. Water exerts pressure
- C. Air has mass
- D. Water occupies space

22. Which one of the following materials allows light to pass but one cannot see through it clearly?

- A. Frosted glass
- B. Mirror
- C. Clean water
- D. Milk

23. Which one of the following is the BEST method of managing plastic waste

- A. Burning
- B. Burying
- C. Recycling

- D. Reusing
24. Which one of the following disease can prevented by wearing gumboots and gloves?
- A. Malaria
 - B. Typhoid
 - C. Bilharzias
 - D. Cholera
25. Which one of the following processes is an advantage to the soil
- A. Ploughing across the contours
 - B. Disposal of vegetable refuses
 - C. Burning of crops remains
 - D. Use of herbicides
26. Which one of the following DOES NOT explain why an improved jiko conserves energy? It has
- A. A cone shape
 - B. Walls made of clay
 - C. A mall space for holding charcoal
 - D. An outer cover made of metals
27. In which of the following is work made easier by changing the direction of force applied?
- A. Ladder
 - B. Flag post
 - C. Staircase
 - D. A road winding up a hill
28. Which one of the following group of foodstuffs constitutes a balanced diet
- A. Potatoes, beans, eggs
 - B. Chicken, cabbage, beans
 - C. Rice, chapatti, beef
 - D. Peas, ugali, spinach
29. The diagram below represents a windsock constructed by pupils
A suitable materials that can used to make the part labeled Q is
- A. Sheet of metal
 - B. Manila paper
 - C. Polythene sheet
 - D. Piece of mosquito net
30. In which one of the following circuit diagrams will the bulb not light
31. Which one of the following is a way of controlling water pollution?
- A. Recycling sewage water
 - B. Reusing domestic water for irrigation
 - C. Storing water in dams
 - D. Using water sparingly

32. The MAIN reason why a baby needs to be breastfed by the mother is that the milk produced
- A. Help to build the baby's immunity
 - B. Help the baby to grow faster
 - C. Help the baby's bones to be strong
 - D. Is easily digested
33. The following are features of a certain type of clouds
- i) White in colour
 - ii) Appear like bundles of cotton
 - iii) Dark grey in colour
 - iv) Mountainous in shape
- Which pair of features are for nimbus clouds
- A. (i) and (ii)
 - B. (i) and (iv)
 - C. (ii) and (iii)
 - D. (iii) and (iv)
34. The diagram below represents a set up that can be used to demonstrate certain processes involved in change of state of matter
- The process that take place at point labeled Q is
- A. Boiling
 - B. Condensation
 - C. Evaporation
 - D. Freezing
35. Which one of the following activities can be used to demonstrate rill erosion?
- A. At a different position of slanting soil surface
 - B. In channels of soil on slanting surface
 - C. On flat surface of soil
 - D. On flat surface using a tin with many hole
36. Which one of the following pairs consists only of materials that are non magnetic?
- A. Copper and aluminum
 - B. Steel and zinc
 - C. Silver and steel
 - D. Iron and aluminum
37. Which one of the following activities will make aluminum foil that was floating on water sink?
- A. Increasing amount of water
 - B. Rolling the aluminum foil into tube
 - C. Making holes on the aluminum foil
 - D. Crushing the aluminum foil into a ball

38. Which one of the following ways of improving soil fertility will take the longest time for plants to get nutrients
- A. Compost manure
 - B. Compound fertilizer
 - C. Farmyard manure
 - D. Green manure
39. Pupils felt a sample of soil with their fingers. This activity was done to investigate
- A. Soil texture
 - B. Presence of water in the soil
 - C. Soil capacity
 - D. Presence of small animals in the soil
40. Which one of the following is NOT a practice for maintaining simple tools?
- A. Oiling
 - B. Using for right purpose
 - C. Sharpening
 - D. Storing in the safe place
41. Newton is a unit for
- A. Mass
 - B. Force
 - C. Pressure
 - D. Volume
42. Which one of the following pairs consists only animals that have webbed feet?
- A. Turkey and crocodile
 - B. Kingfisher and flamingo
 - C. Hippopotamus and toad
 - D. Frog and duck
43. Corrugated iron sheet can be damaged when they come into contact with
- A. Oil spills
 - B. Herbicides
 - C. Industrial gases
 - D. Dust
44. The MAIN reason why black weed should be uprooted from a vegetable farm when they are still young is
- A. Because it is easy to uproot the weed
 - B. To avoid loss of soil nutrients
 - C. To reduce loss of water
 - D. To control the spread of weed
45. The following is modern method of preserving food?
- A. Canning
 - B. Salting

C. Smoking

D. Drying

46. The following are activities involved in demonstrating that light travels in a straight line

I) Light a candle and placed it at opposite end of cardboards

II) Thread a string through the holes

III) Make a hole at the same level in the three cardboards

IV) Cut three similar squares off

Which one of the following is the correct order of activities?

A. (v) (i) (iv) (ii)

B. (v) (iv) (i) (iii)

C. (v) (iii) (iv) (iii)

D. (v) (iii) (iv) (ii) (i)

47. Which one of the following groups consists only of cash crops

A. Maize, tea, yams

B. Arrow roots, groundnuts, cotton

C. Bananas, Irish potatoes, sorghum

D. Sunflower, sisal, coffee

48. The purpose of using rollers to move a load is to

A. Increase the force required to push the load

B. Reduce the force that hinders movements of the load

C. Make the load lighter

D. Control the direction the direction of the movements of the load

49. Which one of the following is TRUE about heat transfer by convection?

It takes place in

A. Solids only

B. Liquids and gases and gases

C. Gases only

D. Solids and gases

50. The chart below represents a simple classification of liquids

Which liquids are represents by P and Q

P

Q

A. Methylated spirit

milk

B. Engine oil

methylated spirit

C. Cooking oil

milk

D. Turpentine

engine oil